HARC Large Scale Urban Forestry Optimization

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Team Members

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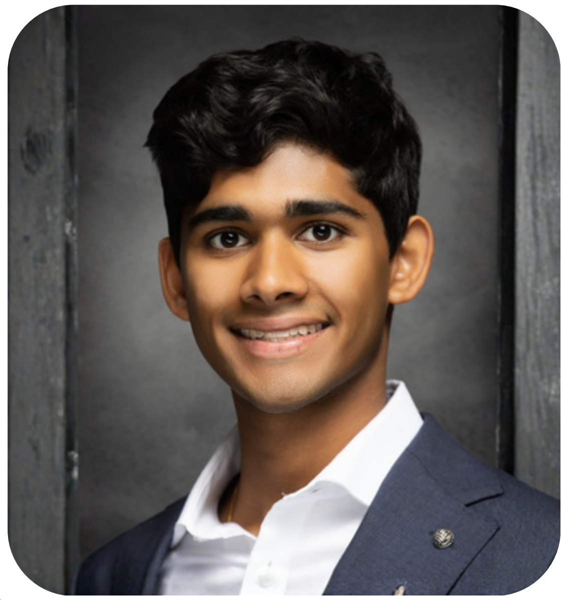

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Optimizer

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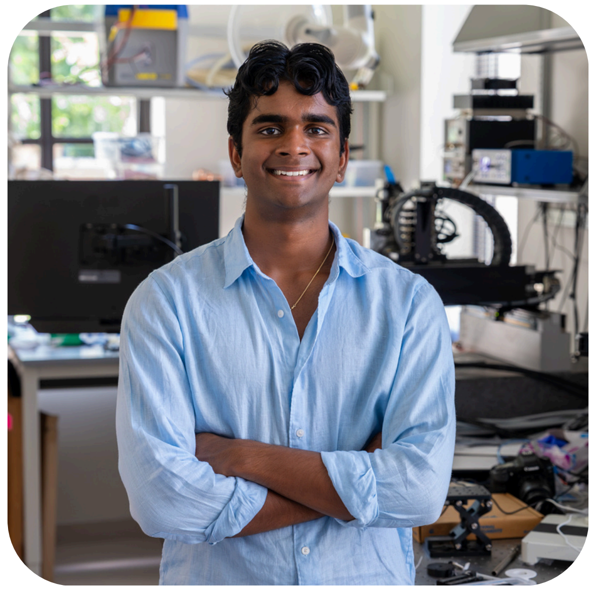

Developer

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Houston Advanced Research Center (HARC)

Dr. Ryan Bare is the Director of Nature-Based Solutions and a Senior Research Scientist at HARC.

He leads multidisciplinary initiatives that integrate hydrology, ecology, and water resource management to address social and environmental challenges across coastal and inland watersheds.

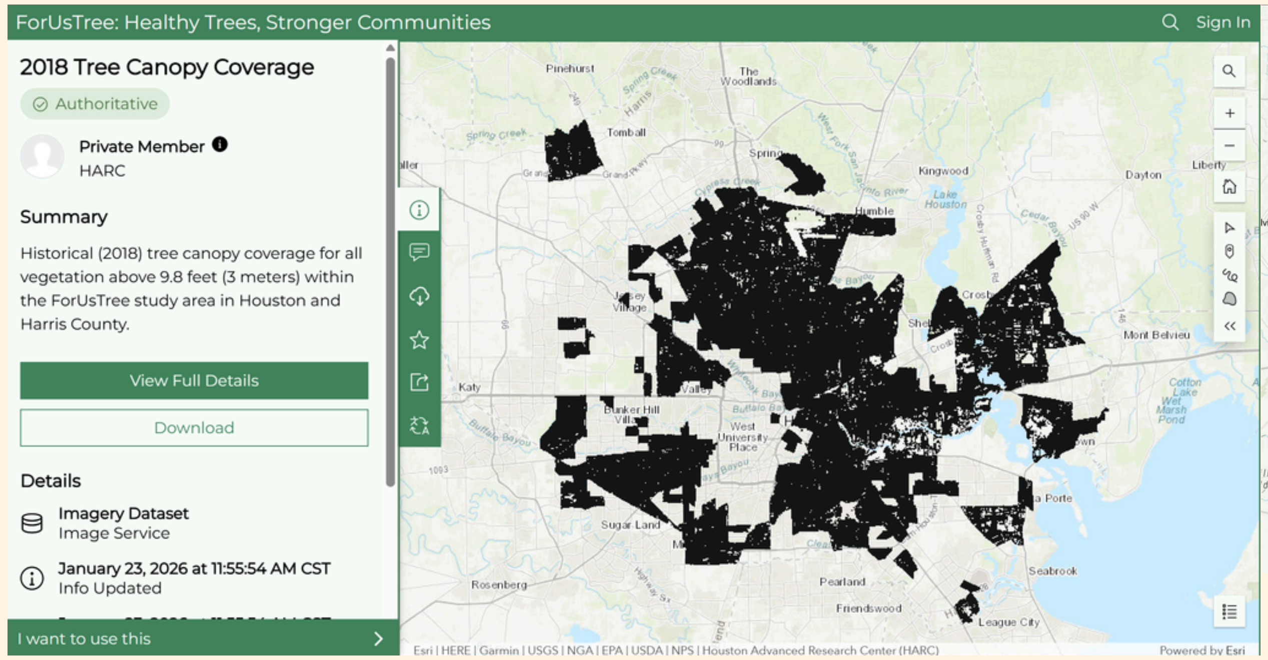
Motivation

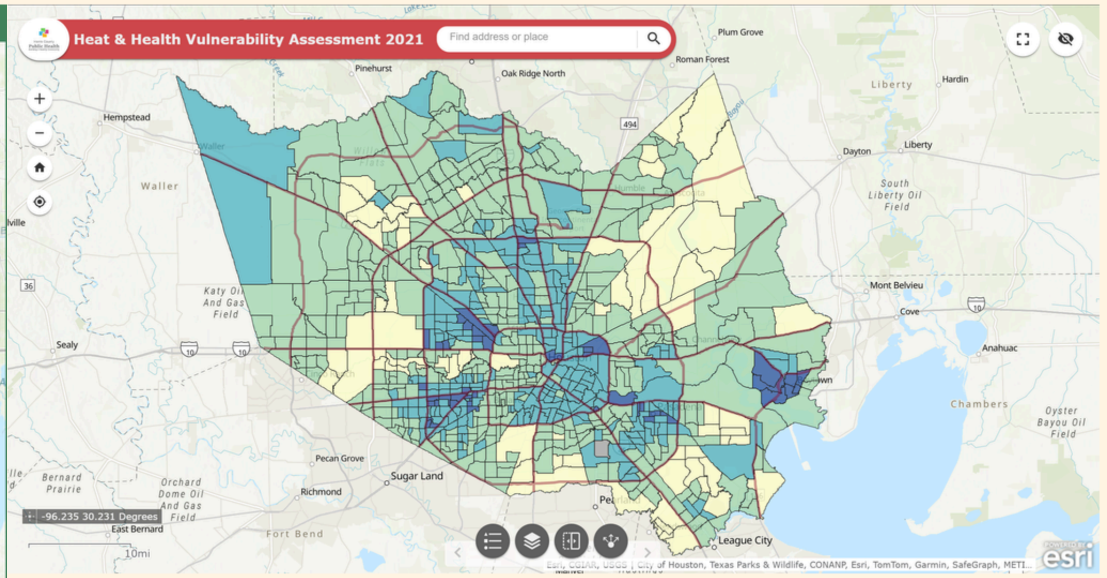
- Houston is a hot city
    - Many health issues come with high heat exposure
- Planting trees provides heat relief
- The challenge: many possible planting locations and tree choices, planting and maintenance costs
- The HARC received a grant and wants to explore the most impactful ways to plant trees

- Task: develop an optimal tree planting plan
- Objective: minimize heat inequality across Houston, subject to budget and geospatial constraints

Data

- Data is geospatial (GIS) captured from LIDAR and is in raster format (a grid of cells [pixels] organized in a matrix, where each pixel holds a value representing information).
    - Tree canopy coverage
    - Tree canopy height
    - Tree canopy density
    - Heat exposure (HARC Afternoon Heat Watch Data)





Problems We Faced

- The original raster data had a spatial resolution of 2 m × 2 m

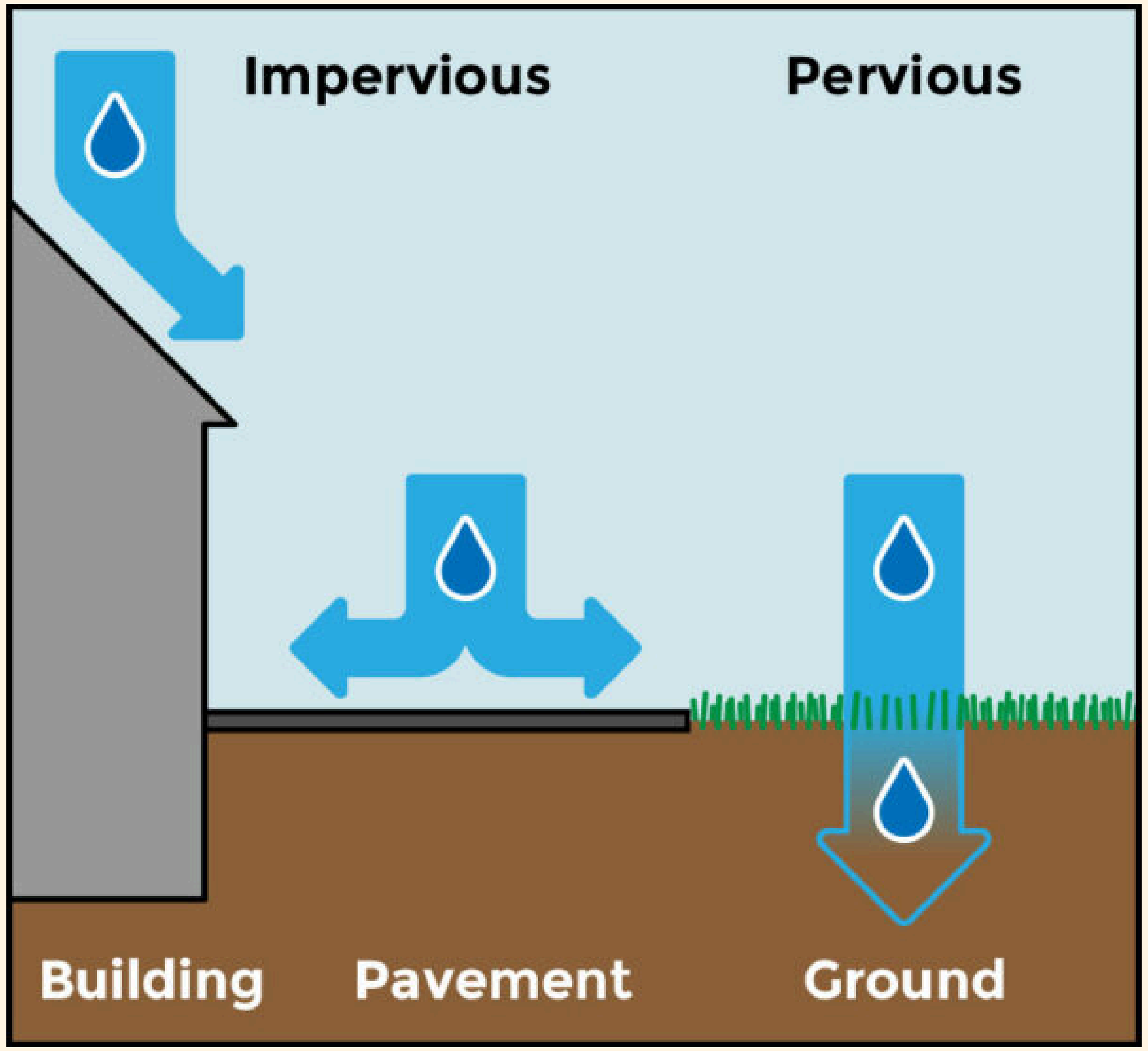
- 2 m x 2 m → 50 m × 50 m
  - Reduction decreased the number of decision variables per tree from approximately 1 billion to 1 million
    - Reduced computational complexity
    - Significantly improved memory usage and runtime
    - Allowed the optimization problem to be solved efficiently

- No data on where we are allowed to plant trees (avoiding recommending parking lots)

How We Used the Impervious Data

- **What is imperviousness?**
  - Imperviousness = the percentage of land covered by surfaces that water cannot soak through
  - Range:
    - 0% → fully grass/soil (plantable)
    - 100% → fully paved (not plantable)
  - Each 50 m × 50 m grid cell has a value between 0 and 1
- **Why it matters for urban heat?**
  - Impervious surfaces:
    - Absorb and retain heat
    - Reduce evapotranspiration
    - Increase surface temperature
  - Higher imperviousness → hotter microclimates

- **Big picture effect - imperviousness affects:**
  - Where planting is allowed
  - How much cooling benefit is generated
  - Which areas are prioritized under budget constraints



How Imperviousness Enters the Model

- <u>How We Used It in the Model</u>
  - **To determine where trees can be planted**
    - We restrict planting in highly impervious areas:

$$a_{ij} = \begin{cases} 1 & q_{ij} \leq \text{threshold} \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ij} \leq a_{ij}$$

    - **Meaning:**
      - Highly paved cells cannot be activated
      - Prevents unrealistic planting in highways or fully built zones
  - **To scale cooling benefit**
    - **Imperviousness modifies marginal cooling impact**

$$\beta_{ij} = 1 + 0.75q_{ij}$$

    - **Objective function contribution:**

$$\max \sum_{i,j} \left(\beta_{ij} h_{ij}^{(1)} z_{ij}^{(1)} + \beta_{ij} h_{ij}^{(2)} z_{ij}^{(2)} \right)$$

    - **Meaning:**
      - Trees planted in dense urban areas generate a larger modeled cooling impact
      - The optimizer prioritizes high-heat built environments

Implementation

- For the implementation, we used Python, specifically the Pyomo modeling framework with the HiGHS solver.

- The model includes approximately 250,000 to 1,000,000 decision variables per tree type, making it a large-scale optimization problem.

Model

Mixed Integer Program

- **Discrete planting decisions:**
  - Tree counts are integers
- **Fixed-charge structure:**
  - Planting at a site incurs a fixed activation cost
- **Inventory limitations:**
  - Each tree type has a global availability cap derived from supplier's inventory
- **Spatial feasibility**:
  - Planting is prohibited in highly impervious cells
- **Piecewise-linear benefits**:
  - Based on existing literature, we assign one level of benefit to canopy coverage of up to 40% and another for canopy gains between 40% and 80%
- **Flexible frontend constraints:**
  - Users may require minimum, maximum, exact, or type-specific planting totals in selected regions

Variables

For each cell (i, j):

- $c_{ij} \in [0, 1]$ → initial canopy fraction
- $q_{ij} \in [0, 1]$ → imperviousness fraction
- $a_{ij} \in \{0, 1\}$ → site feasibility indicator
- $u_{ij} = \max\{0.40 - c_{ij}, 0\}$
- $m_{ij} = \max\{0.80 - c_{ij}, 0\}$

For each tree type:

- $t_k > 0$ → cost per tree of type k,
- $\gamma_k > 0$ → canopy gain contributed by one tree of type k,
- $L_k \geq 0$ → inventory limit for tree type k,
- $\alpha_k > 0$ → capacity usage of tree type k.

Budget and capacity:

- $B > 0$ total planting budget,
- $f > 0$ fixed cost to activate a site,
- $M > 0$ maximum planting capacity per activated site.

Decision variables:

For each cell:

- x_{ijk} denotes the number of trees of type k planted at cell (i, j)
- $y_{ij} \in \{0, 1\}$ equals 1 if site (i, j) is activated,
- $A_{ij} \geq 0$ total added canopy at cell (i, j),
- $z_{(1)ij} \geq 0$ portion of added canopy in the first segment,
- $z_{(2)ij} \geq 0$ portion of added canopy in the second segment,
- $w_{ij} \in \{0, 1\}$ binary variable indicating crossing into segment 2.

Constraints

$$\max \quad 100 \sum_{i \in I} \sum_{j \in J} \left(\beta_{ij} h_{ij}^{(1)} z_{ij}^{(1)} + \beta_{ij} h_{ij}^{(2)} z_{ij}^{(2)} \right)$$

The model maximizes total cooling benefit

$$\text{s.t.} \quad \sum_{k \in K} t_k \sum_{i \in I} \sum_{j \in J} x_{ijk} + f \sum_{i \in I} \sum_{j \in J} y_{ij} \leq B,$$

Budget constraint

$$\sum_{k \in K} \alpha_k x_{ijk} \leq M y_{ij} \qquad \forall (i,j),$$

Per-site capacity constraint

$$y_{ij} \leq a_{ij} \qquad \forall (i,j),$$

Land-feasability constraints

$$\left.\begin{aligned} & A_{ij} = \sum_{k \in K} \gamma_k x_{ijk} \qquad \forall (i,j), \\ & A_{ij} \leq m_{ij} \qquad \forall (i,j), \end{aligned}\right\}$$

Canopy-accounting constraints

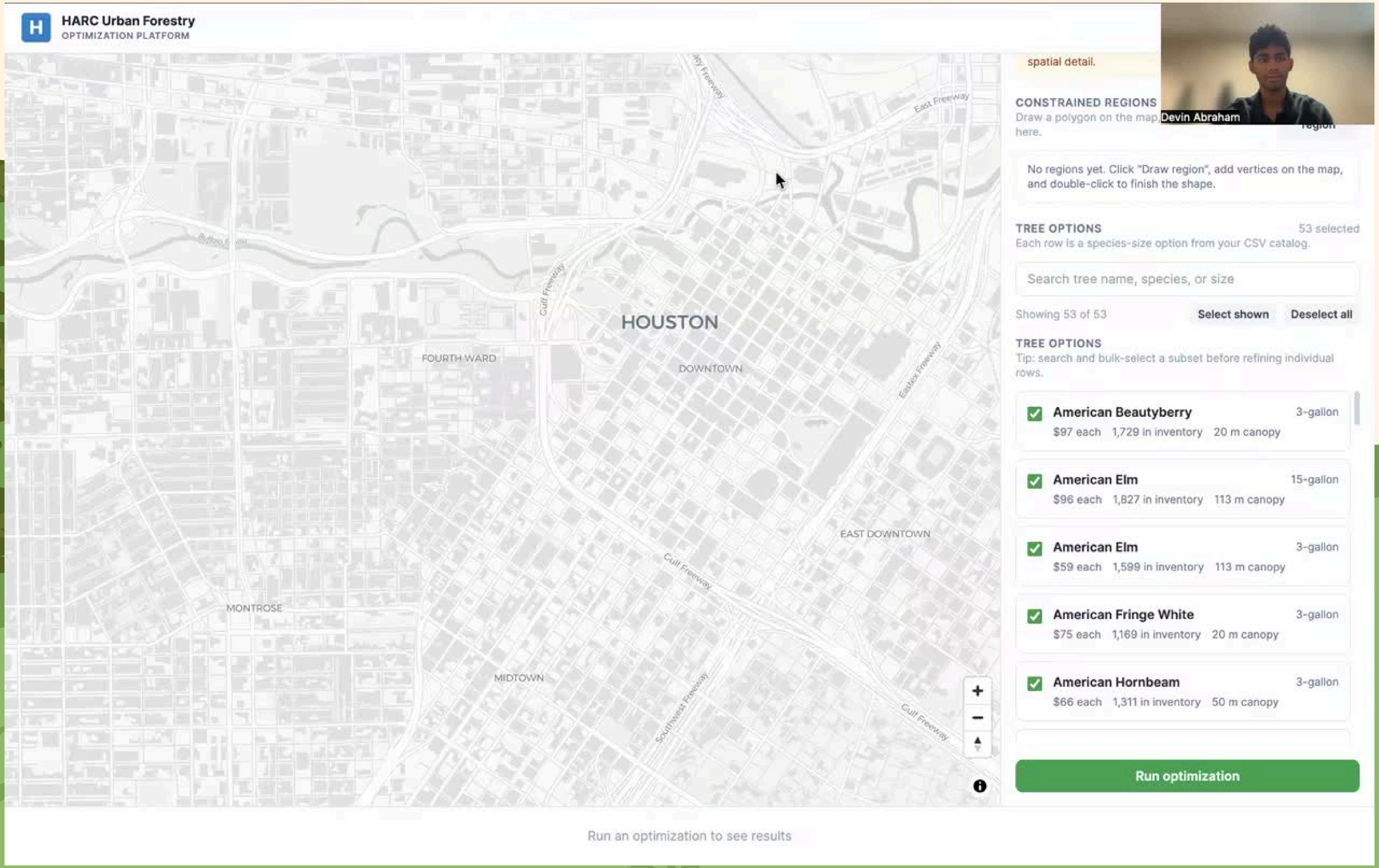
$$\left.\begin{aligned} & z_{ij}^{(1)} + z_{ij}^{(2)} = A_{ij} \qquad \forall (i,j), \\ & z_{ij}^{(1)} \leq u_{ij} \qquad \forall (i,j), \\ & A_{ij} \leq u_{ij} + m_{ij} w_{ij} \qquad \forall (i,j), \\ & z_{ij}^{(2)} \leq m_{ij} w_{ij} \qquad \forall (i,j), \\ & z_{ij}^{(1)} \geq u_{ij} w_{ij} \qquad \forall (i,j), \\ & z_{ij}^{(1)} \leq A_{ij} \qquad \forall (i,j), \end{aligned}\right\}$$

Piecewise Linearization Constraints

$$\sum_{i \in I} \sum_{j \in J} x_{ijk} \leq L_k \qquad \forall k \in K,$$

Tree Inventory Constraints

Demo Video!



H
HARC Urban Forestry
OPTIMIZATION PLATFORM
spatial detail.
CONSTRAINED REGIONS
Draw a polygon on the map
here.
Devin Abraham
No regions yet. Click "Draw region", add vertices on the map, and double-click to finish the shape.
TREE OPTIONS
53 selected
Each row is a species-size option from your CSV catalog.
Search tree name, species, or size
Showing 53 of 53
Select shown
Deselect all
TREE OPTIONS
Tip: search and bulk-select a subset before refining individual rows.
American Beautyberry 3-gallon
$97 each 1,729 in inventory 20 m canopy
American Elm 15-gallon
$96 each 1,827 in inventory 113 m canopy
American Elm 3-gallon
$59 each 1,599 in inventory 113 m canopy
American Fringe White 3-gallon
$75 each 1,169 in inventory 20 m canopy
American Hornbeam 3-gallon
$66 each 1,311 in inventory 50 m canopy
Run optimization
HOUSTON
FOURTH WARD
DOWNTOWN
EAST DOWNTOWN
MONTROSE
MIDTOWN
East Freeway
Gulf Freeway
Eastex Freeway
Southwest Freeway
Run an optimization to see results

Results / Insights

- Prioritizes moderately to highly impervious areas where planting is still feasible, since these locations experience greater cooling impact per unit of added canopy
- Trees are clustered in specific areas rather than spread evenly, driven by fixed site activation costs and the increased benefit of reaching higher canopy coverage
- Areas are pushed past the 40% canopy threshold to unlock higher marginal cooling benefits, with further investment focused on the most valuable locations
- Resources are allocated efficiently by favoring tree types with the highest canopy gain-to-cost ratio, maximizing cooling impact per dollar

Future Work

- Ways to speed up the optimization process
- Increase accuracy in tree planting locations
- Introduce socioeconomic status and inequity between neighborhoods as a factor in the model
- Implement specific tree planting strategies such as tree planting corridors, canopy buffers over impervious areas, etc.

Thank You!